PNVCL-PEGMA Nanohydrogels with Tailored Transition Temperature for Controlled Delivery of 5-Fluorouracil

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ABSTRACT: Sensitive nanohydrogels were prepared via surfactant free emulsion copolymerization of *N*-vinylcaprolactam and poly(ethylene glycol) methyl ether methacrylate, and either *N*-vinylpyrrolidone (VP) or 2-methacryloyloxybenzoic acid (2MBA) to adjust the transition temperature (T_{tr}). The cross-linker ethylene glycol dimethacrylate was used for the polymer network construction. The resulting nanohydrogel sizes are between 120 and 300 nm. ρ-Parameter, obtained from light scattering studies, suggests that core-sell nanogels of flexible chains were obtained. T_{tr} increases with increasing comonomer content (VP or 2MBA) and decreases with decreasing pH for 2MBA containing nanohydrogels. Nanohydrogels containing 15.5% of 2MBA exhibit T_{tr} close to 38 °C. Nanogels are able to control the release of the loaded antineoplastic drug 5-fluorouracil. For the prepared T/pH-sensitive nanogels, the release is slower at pH 7.4 and 37 °C than at tumor conditions: pH 6 and 40 °C. Mathematical models were applied to evaluate the kinetics of drug release; Peppas model fitted best indicating a Fickian diffusion trough a sphere. © 2015 Wiley Periodicals, Inc. J. Polym. Sci., Part A: Polym. Chem. **2015**, *53*, 2662–2672

KEYWORDS: drug delivery systems; nanohydrogels; polylactams; stimuli-sensitive polymers; surfactant free emulsion polymerization

INTRODUCTION Cancer is a major public health concern around the world.[1] In recent decades, the maturation of chemotherapy, by using nanoparticles as controlled delivery systems of anticancer drugs, has been witnessed.[2,3] 5-Fluorouracil (5FU) is an anticancer drug used extensively in cancer chemotherapy for the treatment of solid tumors.[4,5] This drug can be administered via oral or intravenous. However, conventional administration of 5FU has two main disadvantages: first, the production of severe systemic side-effects due to its cytotoxic effect on normal cells; it acts mainly as a thymidylate synthase inhibitor, interferes with nucleic acid synthesis, inhibits DNA synthesis, and eventually halts cell growth;[6] and second, short biological half-life due to rapid metabolism, incomplete and nonuniform oral absorption due to quick degradation by dihydropyramidine dehydrogenase.[7]

In order to substantially improve the administration of antineoplastics, nanocarriers based on responsive polymers have been proposed as highly specific agents that allow controlled drug release.[8,9] The size and shape of nanocarriers are two important parameters that determine the *in vivo* distribution, toxicity, and targeting ability. Additionally, they can influence drug loading, drug release, and *in vitro* and *in vivo* stability. For example, smaller particles have a greater risk of aggregation during storage and incubation *in vitro*, but typically have a longer circulation half-life *in vivo*. The degradation of polymer nanoparticles can be strongly affected by their size as a result of water availability and removal of degradation products.[2,10] Thus, various polymeric systems with typical dimensions in the nanometric range have been tested, as example: micelles,[11,12] stars,[13,14] and hydrogels.[15,16] Particularly, nanohydrogels are swollen polymer networks with unique physicochemical properties in aqueous environment, such as good stability in a broad range of pH and temperatures, good degree of flexibility, and high volume for bioactive drug encapsulation.[17,18] Nanohydrogels from poly(*N*-vinylcaprolactam) (PNVCL) may become powerful anticancer drug carriers given that they are temperature-responsive, very stable against hydrolysis and biocompatible.[19] This

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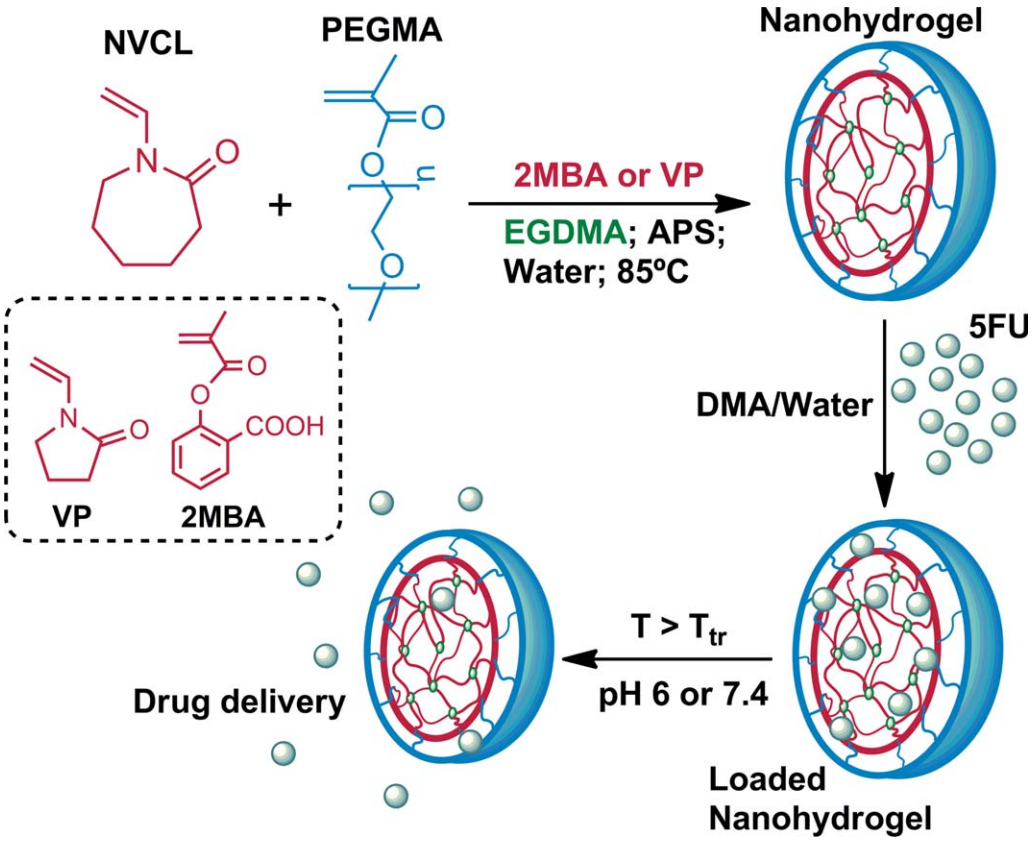

SCHEME 1 Core-shell nanohydrogels prepared by SFEP for controlled delivery of 5FU. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

polymers exhibit a lower critical solution temperature (LCST) near to 32 °C. PNVCL shows a "classical" Flory-Huggins thermoresponsive behavior in water (Type I), therefore, its LCST value decreases with increasing polymer chain length and concentration.[20] The transition temperature (T_{tr}) can be tuned to human body conditions by copolymerization of NVCL with hydrophilic comonomers. In fact, comonomers containing weak carboxylic acids as substituents regulate the temperature-responsive behavior and provide the pH-sensitive character of polymers.[6,21] In the case of nanogels, poly(ethylene glycol) (PEG) helps in tuning their physicochemical properties.[22–24] It is well known that PEG possesses good water solubility, non-toxicity, rapid clearance from the body, and also has been approved for biomedical use by the Federal Food and Drug Administration (FDA) of the United States of America. One of its most important properties is that PEG helps nanomaterials avoiding recognition by the immune system.[25] According to previous reports, nanohydrogels from poly(NVCL-co-methacrylic acid)-PEG[26] and PNVCL-poly(acrylamidoglycolic acid)[6] have potential as pharmaceutical carriers of doxorubicin and 5FU, respectively.

The interest to develop efficient drug carriers based on nanohydrogels, with specific morphology and pendant groups, has led to the design of various preparation methods, such as controlled radical polymerization,[27] emulsion polymerization,[28] and irradiation techniques.[25,29] Chemically (covalent) crosslinked nanogels has been successfully prepared by emulsion polymerization, which can be either classical emulsion polymerization[30–32] or surfactant-free emulsion polymerization (SFEP).[22] SFEP is a simple but powerful tool that avoids the use of surfactants (soap), since residual surfactant may represent a disturbance in the controlled drug delivery system and some of them are reported to be toxic. SFEP involves the polymerization of monomers starting from a homogeneous phase. Nucleation takes place by precipitation of macroradicals. As a result, the initially formed polymer nuclei collide and form larger particles as the polymerization proceeds. Then, particles produced are stabilized by the orientation of their own polymer chains, notably the chain ends originating from initiator molecules.[28] However, it is difficult to prepare nanogels by SFEP, given that the initiator residues incorporated into polymer chains are insufficient to stabilize the precursor nanoparticles. This problem can be overcome by using poly(ethylene glycol) methacrylate (PEGMA) as comonomer. PEGMA becomes covalently attached to the precursor polymer chains and improve the colloidal stability, leading to the nanogel size regulation.[22–24] This synthetic strategy has in common with the recently developed polymerization-induced thermal self-assembly (PITSA)[33] the use of a thermoprecipitating growing polymer chain, nevertheless there are differences in the crosslinking mechanism and stabilization of the produced nanoparticles.

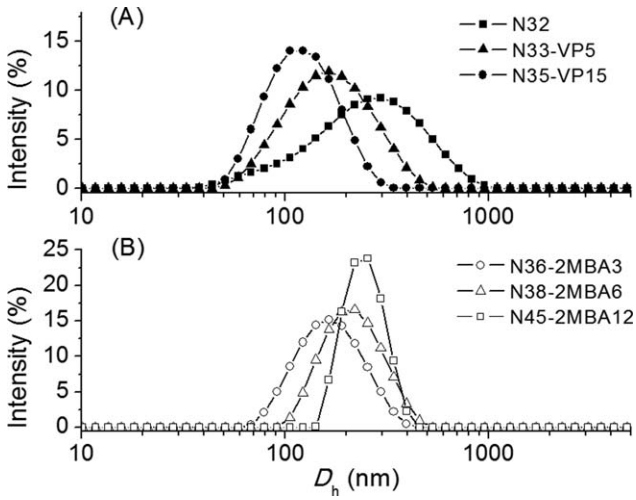

FIGURE 1 Size distribution by intensity of nanogels based on PNVCL-PEGMA containing VP (A) and containing 2MBA (B).

The improvement of chemotherapy by using nanogel-based antineoplastic carriers, with T_{tr} tailored close to human body temperature, is still in demand. In this contribution, responsive nanohydrogels containing mainly PNVCL-PEGMA were prepared by surfactant free emulsion polymerization (Scheme 1). The effect of *N*-vinylpyrrolidone (VP) and 2-methacryloyloxybenzoic acid (2MBA) on particle size and polymer-water interaction was studied. A core-shell morphology is assumed, where the core is a hydrophobic/hydrophilic network, based on NVCL, NVCL-VP, or NVCL-2MBA; while the shell is composed by the hydrophilic PEGMA that stabilizes the drug carrier. Nanohydrogels were tested as carriers of 5FU. Results show the great potential of these materials for their application as biomaterials for controlled drug delivery.

EXPERIMENTAL

Materials

The 2MBA was synthesized following a procedure previously reported.[34] NVCL (98%) was purified by recrystallization in hexane and dried under vacuum prior to use. PEGMA ($M_n = 1100$ g mol^{-1}) and EGDMA (98%) were purified by passing through an inhibitor remover column. VP (>99%) was purified by distillation. Ammonium persulfate (APS, 98%) and 5FU (99%) were used as received. Reagents were purchased from Sigma-Aldrich, México.

Preparation of Nanohydrogels by SFEP

NVCL-PEGMA nanohydrogels were prepared by SFEP method in a Schlenk flask connected with a nitrogen inlet. Ratios of PEGMA, EGDMA, and APS with respect to NVCL in the feed were 45 wt %, 4 mol %, and 4 mol %, respectively. In a typical experiment NVCL (0.275 g, 1.975 mmol), PEGMA (0.225 g, 0.204 mmol), and the crosslinker EGDMA (0.015 g, 0.079 mmol) were dissolved in deionized water (48 mL) at room temperature. The oxygen in the mixture was removed by bubbling nitrogen for 30 min. After that, the monomer solution was heated to 85 °C in a preheated oil bath maintaining constant vigorous stirring. Finally, the thermal initiator, APS (0.018 g, 0.079 mmol), was dissolved in deionized water (2 mL) and added to the reaction system for starting the polymerization. After 30 min, the reaction was stopped by cooling. In the case of nanohydrogels with VP or 2MBA content, the comonomer was added using following percentages with respect to NVCL in the feed: 5, 10, 15 mol % for VP and 3, 6, 9, 12 mol % for 2MBA.

The resulting dispersions were purified via dialysis against deionized water during 5 days, using dialysis membrane, (Spectra/Por®, MWCO = 12,000–14,000), by changing water every day to remove any unreacted monomer and other impurities. Finally, purified nanohydrogels were recovered by freeze drying. The samples were frozen at −4 °C in a conventional freezer for 12 h and then placed into the drying chamber of a Labconco Freeze Dry System Freezone 4.5 (Kansas City, MI), precooled to −50 °C, drying was performed at a pressure of 0.05 mbar for 5 days. Dry samples were stored in refrigerator until its use.

Characterization

Dynamic light scattering (DLS) was used to obtain the size distributions of the nanohydrogels by using a Zetasizer Nano ZS (ZEN3690; Malvern Instruments, Miami, FL) equipped with a red laser (630 nm). The angle of measurement was 90°. Purified and reconstituted samples were analyzed. The freeze dried samples were resuspended to form a 1 wt % dispersion in deionized water under magnetic stirring for 60 min before measurement. The size distribution by intensity using CONTIN analysis is reported. The hydrodynamic diameters (D_h) were calculated using the Stokes–Einstein equation for spheres (eq 1), as usual.[35]

$$R_h = \frac{kT}{6\pi\eta_0 D} \quad (1)$$

where k is the Boltzmann constant, T is the absolute temperature, η_0 is the solvent viscosity, and D is the diffusion coefficient determined from DLS data.

Transition temperature (T_{tr}) and size evolution depending on the comonomer concentration were analyzed by the same Zetasizer equipment. The effect of temperature on the particle size was studied at concentrations of 1 mg mL^{-1} in the range from 20 to 50 °C (equilibration time of 4 min).

Static light scattering (SLS) measurements were carried out to obtain the weight-average molecular weight (M_w), radius of gyration (R_g), and second virial coefficient (A_2). Data were collected using a Goniometer Brookhaven Instruments model BI-200SM-E multiangle DLS/SLS with digital autocorrelator TurboCorr. Dispersions of nanogels were prepared with concentrations of 0.1, 0.15, 0.25, 0.35, and 0.5 mg mL^{-1} in filtered deionized water. Intensity of scattered light was measured for at least five dilutions of each sample in the angular range from 30° to 120°.

Berry plot algorithm was applied for the evaluation of results, using eq 2.[36]

$$\left(\frac{KC}{R_\theta}\right)^{0.5}=\left(\frac{1}{M_w}\right)^{0.5}\left(1+\frac{q^2R_g^2}{6}\right)(1+A_2M_wC) \quad (2)$$

where K is the optical constant ($K=4\pi^2n^2(dn/dC)^2/N_A\lambda_0{}^4$; with n the refractive index of the solvent, dn/dC the refractive index increment of polymer solution, N_A Avogadro's constant, and λ_0 the wavelength of laser light), C is the concentration, R_θ is the normalized scattering intensity (Rayleigh ratio) at the scattering angle θ, M_w is the weight-average molecular weight, R_g is the radius of gyration, A_2 is the second virial coefficient, and q is the magnitude of the scattering vector, expressed as $q = (4\pi n/\lambda_0)\sin(\theta/2)$.

Differential scanning calorimetry (DSC) was used to determine the melting point (T_m) and the glass transition temperature (T_g) of the nanogels. The equipment used was a Modulated DSC, TA Instruments model 2920. Samples were cooled to −10 °C, maintained isothermal for 5 min; afterward the temperature was modulated to +/−0.5 °C every 60 s; and then heated with modulation at a rate of 5 °C min^{-1} to 200 °C in nitrogen atmosphere. Two cycles were run and results reported correspond to the second cycle.

Fourier transform infrared spectroscopy (FTIR) was used to analyze the chemical structure of nanohydrogels using a Spectrum 400, FT-IR/FT-NIR Spectrometer (Perkin Elmer) with a diamond attenuated total reflection (ATR) accessory in the spectral range from 4000 cm^{-1} to 650 cm^{-1}. The samples were used directly as dried powder and 16 scans were collected.

The content of PNVCL in the nanogels was determined by proton nuclear magnetic resonance spectroscopy (^{1}H NMR) by using a Bruker Avance III, 400 MHz equipment. Dried nanogels of known weight (20 mg) were dispersed in deuterated chloroform ($CDCl_3$) (20 mL) by using an ultrasonic bath for 60 min while keeping the dispersion cold (ice bath).

Drug Loading and Encapsulation Efficiency

To demonstrate the 5FU loading and release capacity, the equilibrium swelling method was used. The method was carried out dispersing 20 mg of nanohydrogels in a 5FU solution in dimethylacetamide (DMAc)/Water (90/10 v/v) (10 mg mL^{-1}). DMAc/Water (90/10 v/v) was used due the low solubility of 5FU in pure water. After that, the solution was dialyzed against water for 4 h to remove surface adsorbed 5FU and replace the DMAc by water, using dialysis (membrane: Spectra/Por®, MWCO: 12,000–14,000). The loaded nanohydrogels were frozen at −4 °C in a conventional freezer for 12 h and then placed into the drying chamber of a Labconco Freeze Dry System Freezone 4.5 (Kansas City, MI), precooled to −50 °C, drying was performed at a pressure of 0.05 mbar for 24 h. The drug loaded nanogels were dispersed in a PBS buffer at pH 6 or pH 7.4 and assayed spectrophotometrically for 5FU content using a UV-Vis, Cary 100 spectrophotometer (Varian) at the wavelength at 266 nm. The drug loading (%DL) and encapsulation efficiency (%EE) were calculated using eqs 3 and 4, respectively:[26]

$$\%DL=\left(\frac{MDng}{\text{mass of nanogels}+MDng}\right)\times 100 \quad (3)$$

$$\%EE=\left(\frac{\text{initial MD-MD supernant}}{\text{initial MD}}\right)\times 100 \quad (4)$$

where MD is the mass of drug and MDng is the mass of drug in the nanogel.

In Vitro Release Studies

In vitro released studies were performed under sink conditions in a double jacketed reaction beaker connected to a thermostated water bath at 37 and 40 °C using two different phosphate buffers with 0.1 M ionic strength (pH 6 and 7.4). Typically, 5 mg of 5FU-loaded nanogels were dispersed in 3 mL of the corresponding buffer and then placed into a dialysis bag, which was subsequently immersed in 300 mL of the release medium with moderate stirring at the desired temperature. At predetermined time intervals an aliquot (3 mL) was withdrawn from the delivery medium and assayed by UV-Vis spectroscopy for 5FU; the aliquot was replaced by 3 mL of clean phosphate buffer to keep the volume of the release medium constant. The percentage of drug released (%DR) was calculated by the eq 5.[7]

$$\%DR=\left(\frac{\text{mass of drug released from nanogels}}{\text{mass of total drug loaded in nanogels}}\right)\times 100 \quad (5)$$

RESULTS AND DISCUSSION

Core-Shell Nanohydrogels via SFEP

The synthesis of temperature responsive nanohydrogels via PEGMA-mediated SFEP has been reported previously.[22,24] This synthetic strategy allows the preparation of core-shell nanohydrogels with controlled size, molecular weight, and responsive behavior. For that, a reaction temperature well above the LCST of the corresponding linear polymers is used; at this temperature, the hydrophobic polymer initially formed is stabilized by the hydrophilic comonomer PEGMA, forming a stable nucleus progressively crosslinked, leading to nanogel formation. Since PEGMA contains a polymerizable methacrylate end-group, it is chemically attached to the nanohydrogels forming a PEG shell.[22,24] This strategy to obtain temperature responsive nanohydrogels had some disadvantages in the control of the process, particularly when additional comonomers with different reactivity and different water solubility are used. In this contribution, two families of nanohydrogels were prepared by PEGMA-mediated SFEP. Nanohydrogels were obtained starting with NVCL and PEGMA, adding VP or 2MBA as comonomers, while EGDMA was used as crosslinker. It was envisioned that VP and 2MBA might tune the transition temperature and impart pH-sensitivity behavior to the resulting nanohydrogels. Color of prepared nanohydrogel dispersions was white and

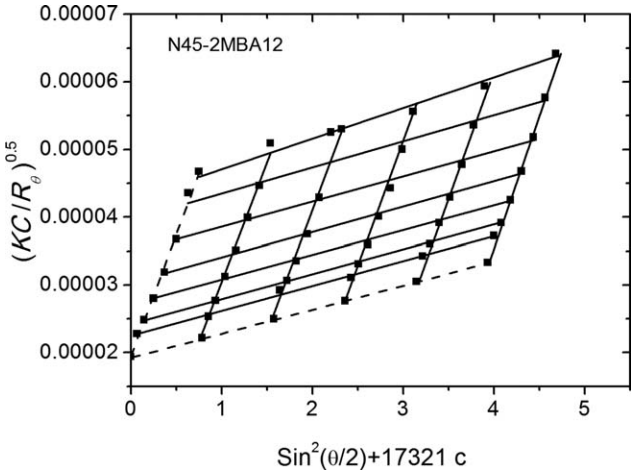

FIGURE 2 Berry plot of nanohydrogel N45-2MBA12.

their intensity decreased as the VP or 2MBA concentration increased. As determined gravimetrically, conversions were around 55 wt %. Colloidal stable particles were characterized thoroughly by means of DLS, SLS, DSC, FTIR, and ^{1}H NMR. The nomenclature, used according to the feed, is the following: PNVCL$_{55\,wt\,\%}$-PEGMA$_{45\,wt\,\%}$ (N32), (PNVCL-VP$_{5\%}$)$_{55\,wt\,\%}$-PEGMA$_{45\,wt\,\%}$ (N33-VP5), (PNVCL-VP$_{10\%}$)$_{55\,wt\,\%}$-PEGMA$_{45\,wt\,\%}$ (N34-VP10), (PNVCL-VP$_{15\%}$)$_{55\,wt\,\%}$-PEGMA$_{45\,wt\,\%}$ (N35-VP15), (PNVCL-2MBA$_{3\%}$)$_{55\,wt\,\%}$-PEGMA$_{45\,wt\,\%}$ (N36-2MBA3), (PNVCL-2MBA$_{6\%}$)$_{55\,wt\,\%}$-PEGMA$_{45\,wt\,\%}$ (N37-2MBA6), (PNVCL-2MBA$_{9\%}$)$_{55\,wt\,\%}$-PEGMA$_{45\,wt\,\%}$ (N38-2MBA9), (PNVCL-2MBA$_{12\%}$)$_{55\,wt\,\%}$-PEGMA$_{45\,wt\,\%}$ (N45-2MBA12).

Figure 1 shows the D_h distributions of nanohydrogels with different content of VP and 2MBA in water. It can be seen that the increase in the VP concentration from 0, 5, 10, and 15 mol %, results in smaller D_h (290, 179, 135, and 128 nm, respectively). Polydispersity index (PDI $= \mu_2/\Gamma^2$, from the cumulant method) decreases with increasing the VP concentration as well. PDI of N32 was found to be 0.37, while PDI of N33-VP5, N34-VP10, and N35-VP15 were 0.26, 0.18, and 0.16, respectively. Narrow size distribution demonstrates homogeneity of NVCL-VP nanogels in aqueous dispersion. The decrease in the particle size may be explained by the fact that, at 85 °C, VP units are less hydrophobic than NVCL ones due the absence of two —CH$_2$— groups on the cyclic ester of the former. The reduction of hydrophobic units leads to good interaction between the initially formed macroradicals and water, and then nanogels are formed starting from smaller stable aggregates of macroradicals.[37] On the other hand, nanohydrogels with higher 2MBA content showed larger particle size (D_h) and lower PDI as compared with the nanohydrogels containing VP: N36-2MBA3 ($D_h \approx 200$ nm, PDI ≈ 0.25), N37-2MBA6 ($D_h \approx 224$ nm, PDI ≈ 0.11), N38-2MBA9 ($D_h \approx 230$ nm, PDI ≈ 0.09), N45-2MBA12 ($D_h \approx 249$ nm, PDI ≈ 0.01). Ionizable acid units intensify the polymer-water interaction which is now pH-dependent. In deionized water (pH $\sim$ 6), 2MBA was found to be more hydrophobic than NVCL since only a part of the acid units are ionized at that pH. The increase in the particle size may be attributed to the higher hydrophobicity of 2MBA units as compared to NVCL ones. The reduction in the number of hydrophilic units, with 2MBA increase, leads to fewer interaction between the initially formed macroradicals and water, and then nanogels are formed starting from larger aggregates of macroradicals.[37] Similar trend was observed in microgels based on *N*-isopropylacrylamide (NIPAAm) with 2MBA prepared by PEGMA-mediated SFEP.[24]

Static light scattering was used to determinate the molecular weight (M_w), radius of gyration (R_g), and the second virial coefficient (A_2) of nanohydrogels. The Berry plot for sample N45-2MBA12, measured in water at 25 °C (Fig. 2), showed linear dependence on the scattering angle, and was related to homogeneous particles (see additional Berry plots in Supporting Information).[37,38] Table 1 summarizes the data of M_w, R_g, and A_2 for six samples. Nanohydrogels show high M_w in the range from 4.67×10^7 to 2.64×10^9 g mol^{-1} and large R_g between 98 and 150 nm. The lowest M_w and R_g correspond to N35-VP15, similar to data obtained for D_h. A clear trend in M_w with increasing 2MBA content in the feed is not observed, while the R_g decreases when the 2MBA in the feed is going from 6 to 12 mol %. Values of A_2 indicate interaction between nanohydrogels and water. The higher the 2MBA content in the polymer network, the lower the A_2 value, indicating the reduction of network-water interactions.

Dynamic and static light scattering were combined to investigate the morphology of nanogels. R_g and R_h were used to calculate the ρ-parameter as $\rho = R_g/R_h$ (Table 1). ρ-values are within the range 0.98–1.53. Values close to 0.98, 1.3, and 1.53 are associated with morphology of "soft sphere," regular

TABLE 1 Light Scattering Data of Nanohydrogels

Nanogel	M_w[a] (g mol^{-1})	R_g[a] (nm)	A_2[a] (mol mL g^{-2})	ρ[b]
N32	1.11 x 10^8	143	2.64 x 10^{-6}	0.98
N35-VP15	4.67 x 10^7	98	1.86 x 10^{-6}	1.53
N36-2MBA3	1.69 x 10^8	115	1.47 x 10^{-6}	1.15
N37-2MBA6	1.18 x 10^9	150	1.24 x 10^{-6}	1.34
N38-2MBA9	1.18 x 10^9	130	1.24 x 10^{-6}	1.13
N45-2MBA12	2.64 x 10^9	128	1.08 x 10^{-6}	1.02

[a] Data obtained in water at 25 °C.

[b] Ratio R_g/R_h.

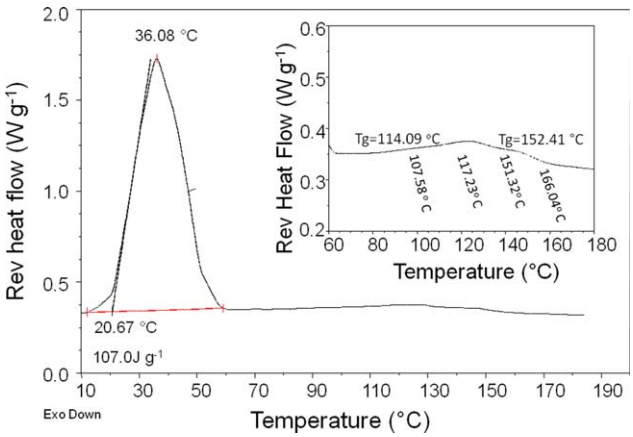

FIGURE 3 DSC thermogram of nanohydrogel N35-VP15. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

stars (uniform arm length) and regular stars (polydisperse arms), respectively.[35,38] Calculated ρ-values suggest that core-shell nanogels of flexible chains were obtained.[37] Sample N32 (PNVCL-PEGMA) has the more compact structure. The flexibility of the polymer network is favored with the incorporation of comonomer at low concentration.

The ρ-parameter decreases when the 2MBA content is higher in the range from 6 to 12 mol %, suggesting that the nanogels become then more compact, resembling again the morphology/flexibility of the N32 nanohydrogel.

The melting point (T_m) and the glass transition (T_g) of dried nanohydrogels were measured by DSC. The presence of different comonomers in the nanohydrogel particles can be confirmed by the characteristic signals of thermal transitions (T_m or T_g) for each comonomer. Figure 3 shows an example of thermogram for nanogel N35-VP15, where three thermal transitions are observed: a sharp crystalline melting transition peak corresponding to PEGMA units ($T_m = 36.08$ °C), a first glass transition ($T_g = 114.09$ °C) corresponding to the crosslinked PNVCL and a second signal of glass transition ($T_g = 152.41$ °C) corresponding to VP. In the case of PNVCL-2MBA-PEGMA (see Supporting Information), the signals for PNVCL and PEGMA are observed; additionally the T_g signal corresponding to the 2MBA is seen close to 170 °C.

The chemical structure of the nanohydrogels was analyzed by FTIR. Figure 4 shows the FTIR spectra of N32, N36-2MBA3, N37-2MBA6, and N45-2MBA12. All spectra contain the typical bands of NVCL units: 1640 cm^{-1} attributed to C═O stretching, and 1500 cm^{-1} corresponding to C—N stretching.[39] The bands at 2880 cm^{-1} and 1440 cm^{-1} are attributed to stretching and bending vibrations of C—H groups, respectively; whereas the bands at 1724 cm^{-1} and 1116 cm^{-1} are attributed to the stretching of carbonyl in the ester and to the C—O stretching in the ether of PEGMA macromonomer units.[40] On the other hand, in spectra of N36-2MBA3, N37-2MBA6, and N45-2MBA12, the signals

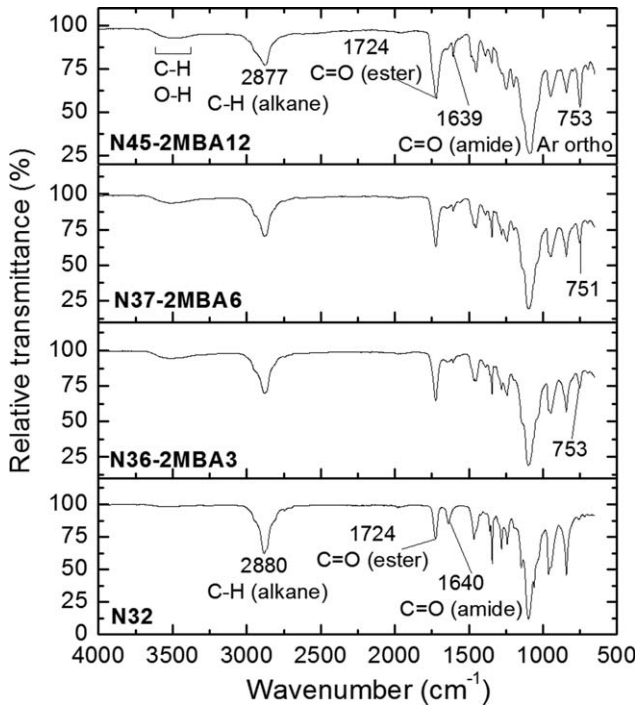

FIGURE 4 FTIR spectra of nanohydrogels from PNVCL-PEGMA (N32) and PNVCL-2MBA-PEGMA.

belonging to NVCL and PEGMA units are present, and additionally, some bands corresponding to 2MBA appeared at 753 cm^{-1} (aromatic ortho substituted) and 3500 cm^{-1} (O—H stretching).[31] The intensity of both bands increases with increasing 2MBA content. In this case, the signal for the stretching of the carbonyl group of 2MBA is superimposed with the carbonyl signal of PEGMA (1724 cm^{-1}) but, the intensity increases with the 2MBA content as well.

The content of PNVCL, PEGMA, and comonomer (VP or 2MBA) in the respective nanogels was determined quantitatively by proton nuclear magnetic resonance (^{1}H NMR). Figure 5 shows the ^{1}H NMR spectrum of N45-2MBA12. The signal at 1.2 ppm is assigned to the —CH_3 groups of the 2MBA and PEGMA (e); the overlapping signal at 1.8 ppm corresponds to the —CH_2—CH_2—CH_2— groups of the lactam ring and polymer backbone (a); the signal at 2.2 ppm corresponds to the two hydrogens of the methylene group (—C(O)—CH_2—) next to carbonyl (d) of lactam and the 2MBA; the signal at 3.2 ppm corresponds to the methylene —N—CH_2— (c) in the lactam ring; the signal at 3.7 pm corresponds to the —CH_2— (f) of PEGMA; the signal at 4.2 ppm corresponds to —CH— of the backbone of polymer PNVCL (b); the signals at 7.5 and 8 ppm are assigned to C—H (aromatic) of 2MBA. The peak at 7.2 ppm corresponds to the deuterated solvent ($CDCl_3$).

The composition was calculated by integration of signals at chemical shifts of 4.2 ppm (NVCL), at chemical shifts of 3.7 ppm (PEGMA) and at chemical shifts 7.5 ppm (2MBA) (see additional spectra in Supporting Information). Table 2

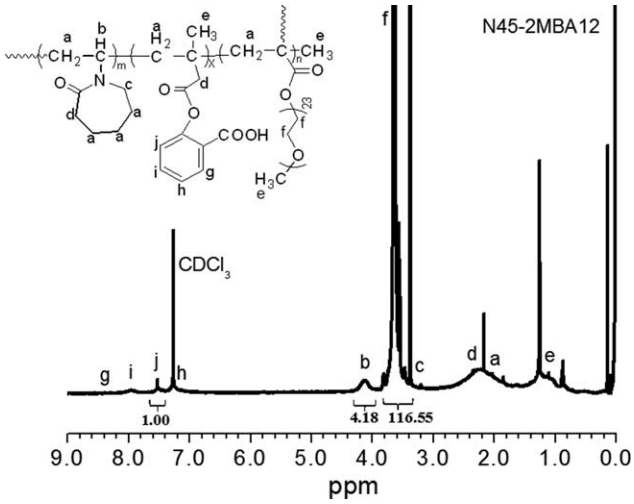

FIGURE 5 ^{1}H NMR spectrum of nanohydrogel N45-2MBA12.

summarizes the compositions of nanoparticles, which prove that NVCL, PEGMA, and comonomer (VP or 2MBA) are present in the respective nanogel structure. In general, NVCL content in nanohydrogels is lower than in the feed, while PEGMA content is higher than expected. The higher the VP or 2MBA in the feed, the higher is the content of these comonomers in the respective nanohydrogels, as expected. Data indicate a preferred incorporation of comonomer over NVCL in the polymer network (see Table 2). The combination of higher content on PEGMA and 2MBA in the nanohydrogels than in the feed is a result of the better polymerizability of methacrylates than *N*-vinyl-lactams under emulsion free radical conditions.

Reported copolymerization parameters for methyl methacrylate are 10 times higher than for NVCL under microemulsion conditions[41] suggesting that methacrylates are preferably incorporated into the copolymer in the initial stages followed by a gradient composition leading eventually to a NVCL rich phase.

The effect of temperature on the nanohydrogel size was analyzed by DLS. Temperature sensitive polymer networks are swollen below the T_{tr} and shrink above it. T_{tr} was estimated from the first derivative of size measured at different temperatures. The effect of the comonomer concentration (VP or 2MBA) on the T_{tr} was studied. Figure 6 shows the T_{tr} of N32 and N33-VP5 in water; and N45-2MBA12 at different pH values (for other samples, see Supporting Information). Nanohydrogel N32 shows a T_{tr} close to 28 °C, while nanohydrogels containing 4.5 (N33-VP5), 20.7 (N34-VP10), and 26.4 mol % (N35-VP15) of VP exhibit T_{tr} of 30, 34, and 42 °C, respectively (Fig. 7 and Table 3). It is noticeable that the width of the transition changes from nanohydrogel to nanohydrogel depending on comonomer and solvent conditions (pH). Since PNVCL is known to show a LCST that is molecular weight dependent,[20] the width of the transition may be also a result of changes in molecular weight between crosslinks for the different nanohydrogels. VP units contribute to overall nanohydrogel hydrophilicity, and to T_{tr} increase, since linear PVP

TABLE 2 Composition of Nanohydrogels Determined by ^{1}H NMR

Nanogel	Monomer	Integration ^{1}H	Feed ratio (mol %)	Content (mol %)
N32	NVCL	1.0	92.2	92
	PEGMA	0.0866	7.8	8
N33-VP5	NVCL	2.79	87.6	70.3
	PEGMA	1.0	7.8	25.2
	VP	0.18	4.6	4.5
N34-VP10	NVCL	3.11	83	60
	PEGMA	1.00	8	19.3
	VP	1.07	9	20.7
N35-VP15	NVCL	2.51	78.4	52.6
	PEGMA	1.00	7.8	21
	VP	1.26	13.8	26.4
N36-2MBA3	NVCL	5.82	89.5	75.7
	PEGMA	0.87	7.7	11.3
	2MBA	1.00	2.8	13
N38-2MBA9	NVCL	4.70	83.9	69.4
	PEGMA	1.07	7.8	15.8
	2MBA	1.00	8.3	14.8
N45-2MBA12	NVCL	4.18	81.2	64.8
	PEGMA	1.27	7.8	19.7
	2MBA	1.00	11	15.5

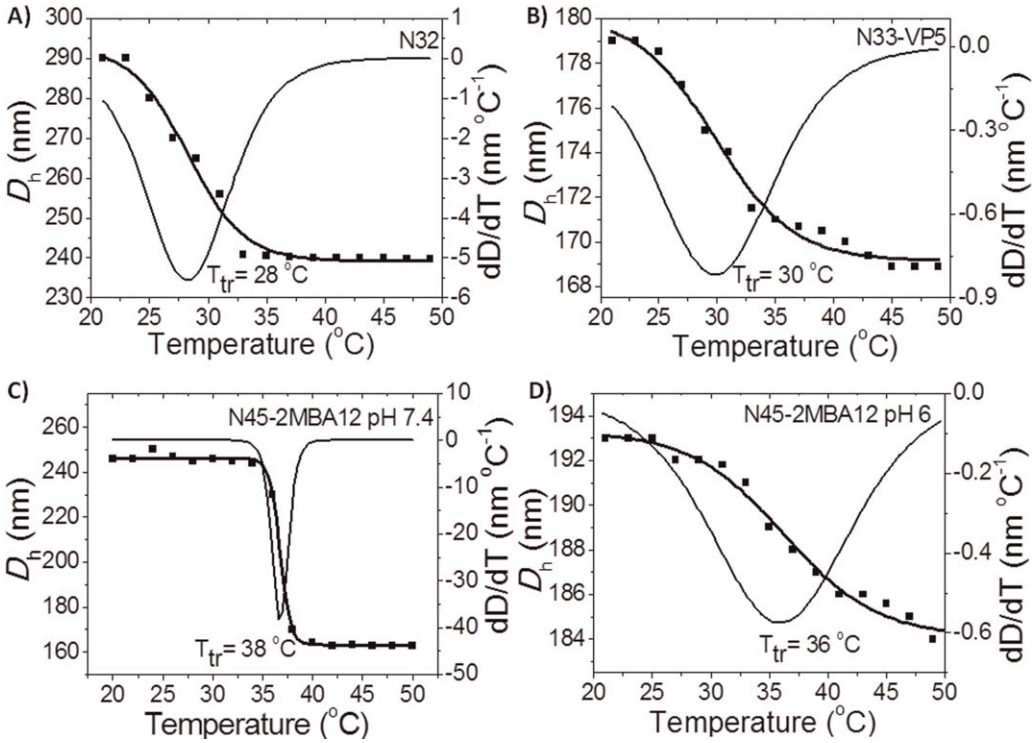

FIGURE 6 D_h of nanogels as function of temperature: a) N32 in water, b) N33-VP5 in water, c) and d) is N45-2MBA12 at pH 7.4 and pH 6, respectively.

shows much higher LCST values (160 °C) as compared to PNVCL (30 °C).[42] Nanohydrogels containing 2MBA exhibit a T/pH-sensitive behavior driven by ionizable acid that change the network-water and intra-network interactions depending on the pH. At pH 7.4, around 95% of 2MBA units are expected to be ionized; however, only 60% of acid units were ionized at pH 6.[43] As an example, the nanohydrogel with 15.5 mol % of 2MBA (N45-2MBA12), studied in buffer solutions at pH 6 and pH 7.4 (Fig. 6), shows at pH 7.4 a T_{tr} close to 38 °C, while at pH 6, the T_{tr} is 36 °C. The T_{tr} shifts to slightly lower values with a decrease in pH, indicating that the interactions within the polymer are getting more preferred to those among the polymer and the solvent molecules.[37] It can be seen, that the T_{tr} increases with the 2MBA concentration as a result of the higher content of ionizable groups (Fig. 7).

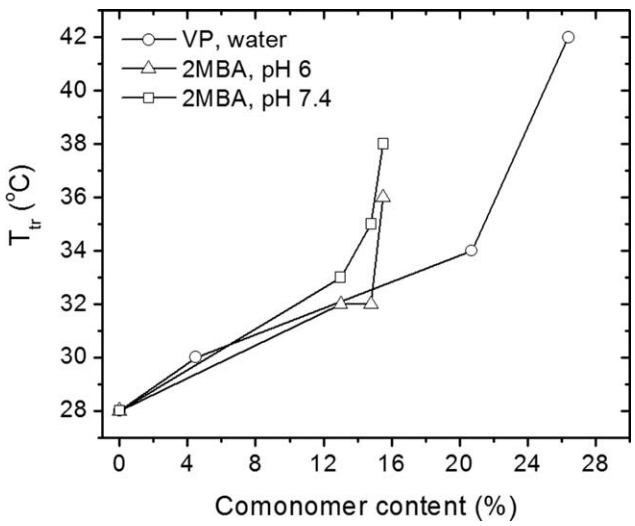

FIGURE 7 T_{tr} of nanohydrogels as function of the comonomer content in the product.

It is noteworthy that the change in T_{tr} occurs within a narrow 2MBA concentration window evidencing the strong effect of an ionizable comonomer into the transition temperature. On the other hand, the swelling of nanogels with acid groups depends on the pH as well (Fig. 6). D_h is larger at pH 7.4, because of the polymer-water interactions and the static repulsion increase in accordance to the dissociation of acid groups and the resultant increment of free ion concentration under alkaline conditions. The incorporation of VP and 2MBA in PEGMA-PNVCL nanogels allows tailoring the T_{tr} of nanocarriers. Nanogels of sample N45-2MBA12 exhibit ideal sensitivity behavior for their application as nanocarriers for controlled drug delivery systems, since its T_{tr} is close to 38 °C at pH 7.4 (no release under normal physiological conditions) and 36 °C at pH 6; thus, the guest drug may be released inside the cells at the endosomal pH.[44]

In Vitro Release Studies of 5FU

The release of drugs from nanohydrogels depends on factors related to the polymer network (composition and network structure), the guest molecule (chemical structure and the molecular weight), and the release conditions (temperature and pH). 5FU is a DNA inhibitor drug widely used in chemotherapy, which has also been entrapped into microgels.[6,7]

TABLE 3 Transition Temperatures of Nanohydrogels at Different Environmental Conditions

Nanogel	Comonomer content	T_{tr} in water	T_{tr} in buffer at pH 7.4	T_{tr} in buffer at pH 6
N32	NVCL: 92%	28 °C	–	–
	PEGMA: 8%			
N33-VP5	NVCL: 70.3%	30 °C	–	–
	PEGMA: 25.2%			
	VP: 4.5%			
N34-VP10	NVCL: 60%	34 °C	–	–
	PEGMA: 19.3%			
	VP: 20.7%			
N35-VP15	NVCL: 52.6%	42 °C	–	–
	PEGMA: 21%			
	VP: 26.4%			
N36-2MBA3	NVCL: 75.7% PEGMA: 11.3% 2MBA: 13%	–	33 °C	32 °C
N38-2MBA9	NVCL: 69.4%	–	35 °C	32 °C
	PEGMA: 15.8%			
	2MBA: 14.8%			
N45-2MBA12	NVCL: 64.8%	–	38 °C	36 °C
	PEGMA: 19.7%			
	2MBA: 15.5%			

With the aim to overcome disadvantages of conventional administration of this drug, the potential of nanohydrogels, based on PNVCL-VP-PEGMA and PNVCL-2MBA-PEGMA, for encapsulation and release of 5FU was tested. The loading content for nanohydrogels N32, N35-VP15, N36-2MBA3, and N45-2MBA12 was 16.6, 17.6, 20.7, and 19.8 wt %, whereas the encapsulation efficiency was 8.1, 8.6, 10.7, and 10.3%, respectively. Nanogels containing 2MBA showed higher drug loading capacity. This is attributed to the presence of hydrogen-bonding and electrostatic forces between 5FU and the polymer network functionalized with carboxylic acid groups.

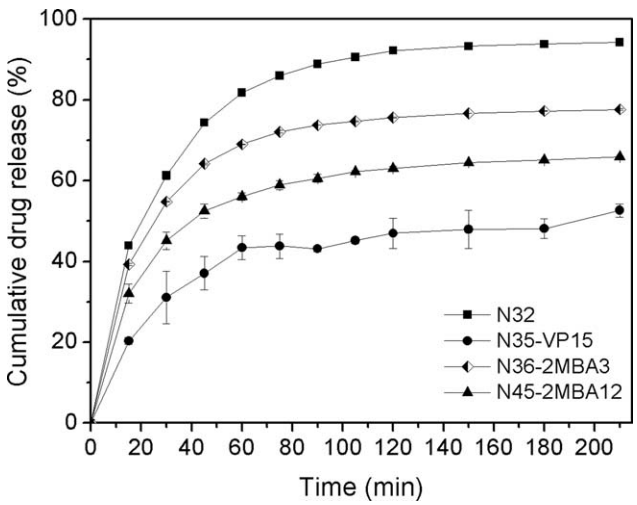

FIGURE 8 Cumulative drug release of 5FU from nanogels based on PNVC-PEGMA, PNVCL-VP-PEGMA, and PNVCL-2MBA-PEGMA at pH 7.4 and 37 °C.

The normal pH of blood for sustaining human life is about 7.4, while tumor tissue has a lower extracellular pH of 6.5 − 7.2 and drops to a pH of 5.0 − 5.5 when reaching the endosomes and lysosomes.[44] Therefore, *in vitro* drug release experiments were carried out imitating physiological conditions of a healthy (37 °C and pH 7.4) and tumor (40 °C and pH 6) tissue. The pH was adjusted using phosphate buffers 0.1 mM. Figure 8 shows the 5FU release profiles of nanohydrogel samples at pH 7.4 and 37 °C. It can be observed, that the drug release profiles depend on the nanogels composition. Drug release from N32 is faster than from nanohydrogels containing comonomers; whereas 5FU release is slower for nanohydrogels with higher T_{tr}, which is consistent with the expected swelling behavior of polymer networks above T_{tr}. This is because it collapses above T_{tr} expelling water with the drug, producing a faster release profile. On the other hand, the drug release decreases with the increase in 2MBA content in the nanohydrogels, because the polymer-drug interactions are favored.

The release rates in nanogels, with VP or 2MBA, at pH 6 and $T = 40$ °C (Fig. 9) are markedly higher as compared to those at pH 7.4 and $T = 37$ °C (Fig. 8). This behavior is attributed to the lower T_{tr} of nanogels at pH 6 as compared to pH 7.4, therefore, the temperature difference to 40 °C is larger.

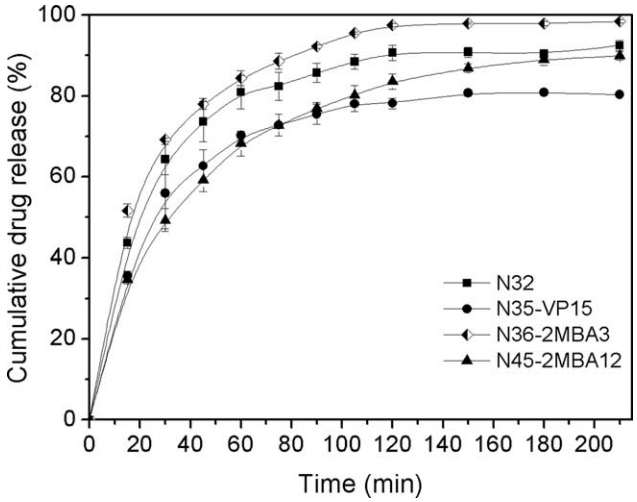

FIGURE 9 Cumulative drug release of 5FU from nanogels based on PNVC-PEGMA, PNVCL-VP-PEGMA, and PNVCL-2MBA-PEGMA at pH 6 and 40 °C.

On the other hand, at pH 6 approximately 60% of 2MBA units are ionized,[42] which decreases the intra-network repulsion as compared to pH 7.4, resulting in a fully collapsed polymer network expelling the 5FU drug even faster.

Nanohydrogel N35-VP15 increases the cumulative 5FU release from 40% at normal physiological conditions (140 min) to 75% at tumor conditions (140 min), nanohydrogel N45-2MBA12 increases the 5FU release from 60 to 85% at the same release time when the same conditions are applied, and so on. The nanohydrogel without comonomer N32 shows the same cumulative release 90% after 140 min at both conditions, demonstrating the importance of the introduction of comonomers to tailor the release behavior of 5FU from these nanohydrogels into tumor tissues.

Mathematical models were applied to evaluate the kinetics of drug release. Data were fitted by eqs (6–8) using linear and no linear regression analysis.[45]

$$\text{Fisrt order}: \quad F = 1 - e^{-kt} \tag{6}$$

$$\text{Higuchi equation}: \quad F = kt^{0.5} \tag{7}$$

$$\text{Peppas equation}: \quad F = kt^{n} \tag{8}$$

where: F = fractional drug release, k = release rate constant for the different equations, n = diffusional exponent.

The results of fitting experimental data to mathematical models are presented in Table 4. The criterion for selecting the most appropriate model was based on linearity (r^2). The best fitting corresponds to Peppa's model. Value of n obtained with this model can be used to judge the drug release mechanism and the matrix geometry;[46] values for n are lower than 0.43, corresponding to a spherical matrix and a drug release mechanism with Fickian diffusion. The values for the release rate constant (k) in all models are higher for the systems at pH 6 and 40 °C as compared to those at pH 7.4 and 37 °C, which confirms the potential of nanogels as carriers of anticancer drugs into tumor tissues.

CONCLUSIONS

Nanohydrogels with a core of PNVCL and a shell of poly(ethyleneglycol) (PEGMA) were successfully synthesized by PEGMA-mediated SFEP. The incorporation of VP and 2MBA allowed adjusting the transition temperature of the materials near to physiological temperature values. The obtained "PEGylated" nanogels possess spherical shape with hydrodynamic diameters within the range of 120–300 nm in their swollen state, representing potentially good candidates for their accumulation in tumors due to the well-known enhanced permeation-retention effect (EPR). The increase of VP concentration leads to smaller nanohydrogels, while the increase of 2MBA content results in bigger nanocarriers. Nanohydrogels with higher content of VP or 2MBA showed transition temperatures near to the physiological value. Furthermore, nanohydrogels were able to encapsulate 5FU with drug loading up to 20.7% due to the electrostatic forces and hydrogen-bond interaction between 5FU and the polymer

TABLE 4 Release Rate for 5FU: *In Vitro* Release Data Fitted with Different Models

Nanogel		First order (k; r^2)[a]	Higuchi (k; r^2)[b]	Peppas (k;n;r^2)[c]
N32	1	0.024;*0.93*	0.098; *0.95*	0.154; 0.401; *0.97*
	2	0.028; *0.96*	0.102; *0.93*	0.165;0.399; *0.96*
N35-VP15	1	0.006; *0.59*	0.049; *0.94*	0.074; 0.408; *0.92*
	2	0.015; *0.78*	0.84; *0.94*	0.137; 0.387; *0.94*
N36-2MBA3	1	0.007; *0.85*	0.036; *0.96*	0.184; 0.309; *0.93*
	2	0.028; *0.96*	0.054; *0.91*	0.242; 0.298; *0.97*
N45-2MBA12	1	0.004; *0.87*	0.031; *0.98*	0.111; 0.315; *0.94*
	2	0.014; *0.97*	0.079; *0.96*	0.147; 0.434; *0.98*

1 and 2 correspond to condition 1 (pH 7.4, 37 °C) and condition 2 (pH 6, 40 °C), respectively.
[a] Calculated by eq 6.
[b] Calculated by eq 7.
[c] Calculated by eq 8.

network. The drug release profiles indicated low extent of release of 5FU from nanohydrogels containing VP and 2MBA at normal physiological conditions (pH 7.4 and 37 °C), and high extent of release from the same nanohydrogels at pH 6 and 40 °C, which represents tumor tissue conditions. Peppas model suggests a Fickian diffusion of 5FU trough spherical nanogels. The *in vitro* release profiles show the potential of the VP and 2MBA modified nanohydrogels as drug delivery systems for oncology therapy.

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REFERENCES AND NOTES

1 R. L. Siegel, K. D. Miller, A. Jemal, *CA Cancer J. Clin*. **2015**, *65*, 5–29.

2 T. Sun, Y. S. Zhang, B. Pang, D. C. Hyun, M. Yang, Y. Xia, *Angew. Chem. Int. Ed*. **2014**, *53*, 12320–12364.

3 B. K. Lee, Y. H. Yun, K. Park, *Chem. Eng. Sci*. **2015**, *125*, 158–164.

4 J. Guo, A.-w. Zhou, Y.-c. Fu, U. N. Verma, D. Tripathy, E. Pfrenkel, C. R. Becerra, *Acta Pharmacol. Sin*. **2006**, *27*, 1375–1381.

5 M. Malet-Martino, R. Martino, *Oncologist* **2002**, *7*, 288–323.

6 K. Madhusudana-Rao, B. Mallikarjuna, K. S. V. Krishna-Rao, S. Siraj, K. Chowdoji-Rao, M. C. S. Subha, *Colloids Surf. B* **2013**, *102*, 891–897.

7 N. S. Rejinold, K. P. Chennazhi, S. V. Nair, H. Tamura, R. Jayakumar, *Carbohydr. Polym*. **2011**, *83*, 776–786.

8 S.-M. Lee, S.-B. T. Nguyen, *Macromolecules* **2013**, *46*, 9169–9180.

9 P. J. Honey, J. Rijo, A. Anju, K. R. Anoop, *Acta Pharm. Sin. B* **2014**, *4*, 120–127.

10 L. Y. Qiu, Y. H. Bae, *Pharm. Res*. **2006**, *23*, 1–30.

11 Y. Ma, X. Jiang, R. Zhuo, *J. Polym. Sci., Part A: Polym. Chem*. **2013**, *51*, 3917–3924.

12 X. Guo, C. Shi, G. Yang, J. Wang, Z. Cai, S. Zhou, *Chem. Mater*. **2014**, *26*, 4405–4418.

13 M. Zhang, Q. Xiong, J. Chen, Y. Wang, Q. Zhang, *Polym. Chem*. **2013**, *4*, 5086–5095.

14 J. Alvarez-Sanchez, A. Licea-Claverie, J. M. Cornejo-Bravo, C. W. Frank, *React. Funct. Polym*. **2011**, *71*, 1077–1088.

15 S. Santos-Pedrosa, C. Goncalves, L. David, M. Gama, *Macromol. Biosci*. **2014**, *14*, 1556–1568.

16 R. Jayakumar, A. Nair, N. S. Rejinold, S. Maya, S.V. Nair, *Carbohydr. Polym*. **2012**, *87*, 2352–2356.

17 A. V. Kabanov, S. V. Vinogradov, *Angew. Chem. Int. Ed*. **2009**, *48*, 5418–5429.

18 Y. Sasaki, K. Akiyoshi, *Chem. Rec*. **2010**, *10*, 366–376.

19 J. Liu, A. Debuigne, C. Detrembleur, C. Jérôme, *Adv. Healthcare Mater*. **2014**, *3*, 1941–1968.

20 J. Ramos, A. Imaz, J. Forcada, *Polym. Chem*. **2012**, *3*, 852–856.

21 S. M. Ponce-Vargas, N. A. Cortez-Lemus, A. Licea-Claveríe, *Macromol. Symp*. **2013**, *325-326*, 56–70.

22 M. A. Gonzalez-Ayon, N. A. Cortez-Lemus, A. Zizumbo-Lopez, A. Licea-Claverie, *Soft Mater*. **2014**, *12*, 315–325.

23 Q. Zhang, L. Zha, J. Ma, B. Liang, *J. Colloid Interface Sci*. **2009**, *330*, 330–336.

24 A. Serrano-Medina, J. M. Cornejo-Bravo, A. Licea-Claveríe, *J. Colloid Interface Sci*. **2012**, *369*, 82–90.

25 W. C. Lee, Y. C. Li, I. M. Chu, *Macromol. Biosci*. **2006**, *6*, 846–854.

26 Y. Wang, J. Nie, B. Chang, Y. Sun, W. Yang, *Biomacromolecules* **2013**, *14*, 3034–3046.

27 N. Sanson, J. Rieger, *Polym. Chem*. **2010**, *1*, 965–977.

28 R. Arshady, *Colloid Polym. Sci*. **1992**, *270*, 717–732.

29 S. Zschoche, J. C. Rueda, M. Binner, H. Komber, A. Janke, K. F. Arndt, S. Lehmann, B. Voit, *Macromol. Chem. Phys*. **2012**, *213*, 215–226.

30 A. Imaz, J. Forcada, *J. Polym. Sci., Part A: Polym. Chem*. **2008**, *46*, 2510–1524.

31 S. F. Medeiros, A. M. Santos, H. Fessi, A. Elaissari, *J. Polym. Sci., Part A: Polym. Chem*. **2010**, *48*, 3932–3941.

32 A. Imaz, J. Forcada, *J. Polym. Sci., Part A: Polym. Chem*. **2010**, *48*, 1173–1181.

33 C. A. Figg, A. Simula, K. A. Gebre, B. S. Tucker, D. M. Haddleton, B. S. Sumerlin, *Chem. Sci*. **2015**, *6*, 1230–1236.

34 A. Licea, E. Rogel, J. A. López, L. A. Castillo, J. M. Cornejo, K. F. Arndt, *Des. Monomers Polym*. **2003**, *6*, 67–80.

35 W. Burchard, *Adv. Polym. Sci*. **1999**, *143*, 113–194.

36 E. He, P. Ravi, K. C. Tam, *Langmuir* **2007**, *23*, 2382–2388.

37 L. A. Picos-Corrales, A. Licea-Claveríe, K. F. Arndt, *React. Funct. Polym*. **2014**, *75*, 31–40.

38 L. A. Picos-Corrales, A. Licea-Claveríe, K. F. Arndt, *J. Polym. Sci., Part A: Polym. Chem*. **2012**, *50*, 4277–4287.

39 S. Cavus, E. Cakal, *Ind. Eng. Chem. Res*. **2012**, *51*, 1218−1226.

40 L. Yuan, W. L. Chen, J. Li, J. Hu, J. Yan, D. Yang, *J. Polym. Sci., Part A: Polym. Chem*. **2012**, *50*, 4579−4588.

41 Z. S. Shah, A. Pal, R. Gude, S. Devi. *Eur. Polym. J*. **2010**, *46*, 958–967.

42 B. Jeong, S. W. Kim, Y. H. Bae, *Adv. Drug Delivery Rev*. **2002**, *54*, 37–51.

43 L. A. Picos-Corrales, A. Licea-Claveríe, J. M. Cornejo-Bravo, S. Schwarz, K. F. Arndt, *Macromol. Chem. Phys*. **2012**, *213*, 301–314.

44 Y. C. Chen, L. C. Liao, P. L. Lu, C. L. Lo, H. C. Tsai, C. Y. Huang, K. C. Wei, T. C. Yen, G. H. Hsiue, *Biomaterials* **2012**, *33*, 4576–4588.

45 S. N. Dobic, J. M. Filipovic, S. L. Tomic, *Chem. Eng. J*. **2012**, *179*, 372–380.

46 J. Siepmann, F. Siepmann, *Int. J. Pharm*. **2008**, *364*, 328–343.